GlucoTwin

An Intelligent Predictive Digital Twin for Proactive Glucose-Risk Monitoring

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- Live Web Application: https://gluco-twin-eta.vercel.app/
- Live Production API: https://gluco-twin.onrender.com
- Open Source Repository: https://github.com/Yuvashri2516/Gluco-Twin
- Demo Video Link: [Demo Video URL Pending Final Recording — See Checklist]
- Notice: Research Prototype Built with Clinically Modeled Synthetic Telemetry

Problem Statement & Clinical Motivation

The Reactive Paradox of Modern Diabetes Management

- Delayed Intervention: Traditional CGM alerts trigger only AFTER acute excursions occur (hypoglycemia < 70 mg/dL or hyperglycemia >= 180 mg/dL).
- Fragmented Data Silos: Electronic health records (EHR labs, genetic markers) rarely communicate with real-time w earable streams.
- Autonomic Blind Spots: Glycemic variability directly impacts cardiac autonomic regulation (HR/HRV), but vital si gns are monitored in isolation.
- Cognitive Burden: Patients and clinicians lack transparent, counterfactual forecasting to understand the forward impact of meals and exercise.

Healthcare Use Case & Target Audience

Proactive Surveillance for Outpatient & Clinical Care Teams

- Endocrinology & Primary Care Teams: Continuous surveillance of high-risk diabetic cohorts with triage prioritization.
- Certified Diabetes Care & Education Specialists (CDCES): Patient education using transparent feature contribution breakdowns.
- Outpatient Remote Monitoring: Early notification of impending glycemic excursions 120 minutes before clinical threshold breaches.
- Strict Non-Prescriptive Scope: GlucoTwin provides surveillance and risk forecasting, never automated insulin dosing or medical prescriptions.

The GlucoTwin Proposed Solution

A Closed-Loop Virtual Twin Fusing Historical EHR & Live Streams

- Multimodal Fusion: Unifies historical EHR (labs, genetics) with dynamic physiological streams (CGM, HR, HRV, sle ep, step activity).
- Predictive Machine Learning: Forward-looking binary classifier predicting spikes (>= 180 mg/dL within 120 min) w ith ROC-AUC 0.825.
- Dual-Source Alert Engine: Explicitly distinguishes observed threshold violations from model-predicted future ris ks.
- Demonstrable Simulation Workflow: Interactive +15m sensor epoch feed updates twin state and triggers live alert triage.
- What-If Counterfactual Sandbox: Educational simulator modeling carbohydrate impact and exercise mitigation.

Digital Twin Concept & Virtual Phenotyping

Continuous State Representation of Patient Metabolic Dynamic

- Static Phenotype Vector: Age, gender, diabetes subtype (T1D/T2D/Pre-diabetic), years diagnosed, weight, height, BMI.
- Dynamic Telemetry Surveillance: 24-hour moving baseline anchored to the latest sensor timestamp.
- 1-Hour Delta Kinetics: Calculates rolling baseline excursions: ΔGlucose, ΔHeartRate, and ΔHRV against 24h baseline.
- Sensor Coverage Integrity: Quantifies reading density and flags telemetry gaps (> 45 min) to prevent spurious predictions.

Multimodal Data Architecture: EHR & Telemetry

Unified Clinical Data Model Across 5 Diverse Cohorts

- Historical EHR: Complete demographic profiles with longitudinal laboratory panels (HbA1c, Fasting Glucose, Lipids).
- Genetic Susceptibility Markers: Simulated risk allele records (TCF7L2 rs7903146, PPARG Pro12Ala, KCNJ11, SLC30A8, KCNQ1).
- High-Frequency Dynamic Streams: Synchronized 15-minute continuous glucose, heart rate (bpm), and HRV (RMSSD).
- Polysomnography & Activity: Sleep architecture stages (Deep, Light, REM, Awake) and interval vs cumulative daily step counts.
- Synthetic Integrity: All synthetic records carry explicit is_synthetic=True database flags and clinical disclaimers.

System Architecture & End-to-End Pipeline

Decoupled Modern Stack: React 19 Frontend + FastAPI Backend + SQLite

- Client Tier: React 19 Single Page Application with Tailwind/Vanilla CSS, Recharts visualizations, and dark medic al aesthetic.
- API Tier: FastAPI asynchronous REST services deployed on Render with strict Pydantic v2 schemas and CORS middlew are.
- Persistence Tier: SQLite database with indexed queries, transactional commits, and unique deduplication constrai nts.
- Standalone Architecture Deliverable: High-resolution vector diagram available in submission/GlucoTwin_Architectu re.pdf.

Multimodal Data Fusion Workflow

Disciplined Clinical & Machine Learning Information Routing

- Clinical Context Stream: Historical diagnoses, lab panels, and genetic markers populate patient profiles and ris k stratification.
- Circadian & Exertion Context: Sleep stages provide restorative baseline; step counts inform diurnal activity and What-If models.
- Model Feature Matrix: 14 leak-free features extracted strictly from t <= T historical lookback windows.
- Scientific Integrity: Labs and genetics are displayed for clinical context but NOT forced into the ML matrix to prevent uncalibrated bias.

Machine Learning Spike Prediction Engine

Predicting Glycemic Excursions (>= 180 mg/dL) 2 Hours Ahead

- Prediction Horizon: 120 minutes forward window strictly excluding readings after prediction timestamp T.
- Model Pipeline: StandardScaler fitted exclusively on training set + LogisticRegression(penalty='l2', C=1.0).
- Leakage Safeguards: 2-hour purged holdout split between training and evaluation partitions prevents temporal dat a contamination.
- Feature Explanations: Model returns exact linear feature contributions and directional risk impacts for each pre diction.

ML Model Evaluation & Benchmark Comparison

Rigorous Evaluation on Held-Out Test Cohorts

- Chronological Holdout (1,114 test samples): ROC-AUC 0.9445 | Accuracy 87.6% | Precision 0.645 | Recall 0.408 | F 1 0.500 | Brier 0.0688 (Baseline partition: ROC-AUC 0.705).
- Unseen-Patient Generalization: Cross-patient evaluation (Leave-One-Patient-Out) yields mean ROC-AUC ~0.5156 acro ss cohorts with spikes.
- Evaluation Split Distinction: Chronological holdout tests forward trajectory forecasting on known profiles; unse en evaluation tests population transfer.
- Personalized vs Population Scope: Model is designed as a personalized patient trajectory predictor, not an uncal ibrated universal screening tool.

Doctor-Facing Dashboard Walkthrough

Unified Clinical Cockpit for Time-Series & Patient Overview

- Cohort Switching: Instant switching between 5 diverse patient personas with dynamic chart updates and zero state leakage.
- Synchronized Multi-Vital Telemetry: Interactive time-series charts aligning CGM curves, heart rate, and autonomic HRV.
- Historical EHR Timeline: Tabular and badge views of longitudinal lab panels and synthetic genetic susceptibility alleles.
- Sleep & Step Analytics: Visual hypnogram sleep stages, efficiency percentages, and 15-minute interval physical activity bars.

Demonstrable Digital Twin Live Update Workflow

Interactive Simulated Sensor Feed Ingestion

- Controlled Ingestion: Clinicians trigger simulated +15m sensor epochs directly from the Digital Twin panel.
- Telemetry Advancement: Generates synchronized glucose, HR, HRV, and step count readings without continuous unpro mpted polling.
- Instant Twin Recomputation: Recalculates 24h baseline vs 1h delta vectors and refreshes dashboard status tags.
- Immediate Alert Triggering: Automatically evaluates clinical thresholds on incoming telemetry and updates notifi cation badges.

Clinical Alert Surveillance & Simulated Pager

Two-Tier Alert Categorization with Clinician Triage Routing

- Observed Telemetry Alerts: Triggered on actual breaches: Severe Hypo (<54 mg/dL), Hyper (>=180 mg/dL), Severe Hyper (>=250 mg/dL).
- Model-Predicted Alerts: Purple brain badge alerts for upcoming 2-hour predicted spikes with risk score >= 0.50.
- Lifecycle Workflow: Full Active -> Acknowledged -> Resolved state transitions with timestamps and deduplication keys.
- Simulated Clinician Pager: Demonstrates hospital triage queue routing with explicit disclaimer (External SMS/Email unconfigured).

What-If Counterfactual Sandbox

Educational Simulation of Meal Carbs & Exercise Impact

- Single-Compartment Linear Heuristic: ΔG = Carbs * (0.40 * 70/weight * K_type) - Activity * 0.75 mg/dL/min.
- Subtype Sensitivity Factors: Tailored factors for Type 1 (1.30), Type 2 (1.15), and Pre-diabetic (0.90) cohorts.
- Mitigation Testing: Demonstrates how 30 minutes of physical activity blunts postprandial glycemic excursions.
- Safety Prominence: Prominently watermarked 'SIMULATED — NOT A CLINICAL PREDICTION' with medical disclaimer.

Automated Testing & Quality Verification

100% Passing Automated Test Suites & Production Build Verification

- Backend Test Suite: 71 automated pytest unit and integration tests passing with 100% success rate.
- Database Isolation: All automated tests execute on isolated in-memory SQLite (sqlite:///:memory:) protecting per sistent data.
- Comprehensive Coverage: Tests cover EHR endpoints, lab/genetics, sleep/steps, alert lifecycle, and ML leakage pr evention.
- Frontend Verification: ESLint passed with 0 errors; Vite production bundle built in 942ms without compilation is sues.

Deployment Architecture & Production Health

Continuous Deployment on Vercel & Render

- Frontend Deployed: Vercel serverless edge network (https://gluco-twin-eta.vercel.app/) with automated production builds.
- Backend Deployed: Render web service container (https://gluco-twin.onrender.com) with automated health probes.
- Environment Security: All deployment secrets, API tokens, and credentials are protected; zero sensitive leaks in Git.
- CORS & Health Probes: Configured CORS for production domains and live /api/health monitoring endpoint.

Known Limitations & Responsible Disclaimers

Honest Engineering Boundaries for a Research Proof-of-Concept

- Synthetic Data Foundation: Clinically simulated physiological models capture diurnal patterns but cannot replica te complex biological pharmacokinetics.
- Unseen-Patient Generalization Boundary: Cross-patient ROC-AUC is ~0.52 on 5 personas; model requires individual historical trajectory data to forecast.
- No Medical Device Claims: GlucoTwin is an academic engineering prototype and must NOT be used for clinical diagn osis or insulin dosing.
- Local Notification Demonstration: Hospital triage pagers are simulated locally; external SMS and email are inten tionally unconfigured.

Future Roadmap: Post-Hackathon Improvements

Path Toward Clinical Validation & Commercial Device Integration

- Hardware Wearable Integration: Native Bluetooth Low Energy (BLE) ingestion from Dexcom G7, FreeStyle Libre 3, an d Apple Watch.
- Pharmacokinetic Modeling: Differential equation modeling for insulin-on-board (IOB) decay and gastric carbohydra te absorption.
- Deep Learning Architectures: Recurrent LSTM / Transformer models leveraging multi-week longitudinal telemetry.
- Hospital EHR Interoperability: HL7 FHIR API connectors (SMART on FHIR) for Epic and Cerner clinical EHR integrat ion.

Conclusion & Hackathon Submission Summary

A Fully Realized Multimodal Digital Twin Proof-of-Concept

- Deliverables Package: Standalone Architecture Diagram (PDF/SVG), Presentation Deck (PPTX/PDF), and 20-min Demo S cript.
- Proven Reliability: 71 backend tests passed, frontend lint clean, production build verified, and database 100% p reserved.
- Holistic Healthcare Vision: Demonstrates the power of continuous digital twins to transform reactive medicine in to proactive health.
- Submission Artifacts: Available in root /submission folder for comprehensive evaluator review.